MAST20026 REAL ANALYSIS

Limit Points, Convergence, and Divergence – Solutions

7A

A Note on the Solutions: This solution sheet is intended for you verify correctness of your work during the tutorial. The solutions here do not necessarily meet the standards for exposition expected on assignment submissions.

1. **Limit Points.** Find all limit points of each set:

   (a) $[1, 2]$

   (b) $[1, 2]$

   (c) $\{\}$

   (d) $\mathbb{R}$

   (e) $(-\infty, -1] \cup [1, \infty)$

   (f) $\mathbb{R}$

   (g) $\{0\}$

2. $\epsilon - \delta$ **Limits.** Using an $\epsilon - \delta$ argument, prove that each of the statements below are true. For help on (e) and (f), look back at *Test Your Understanding* in 5.1.

   (a) Any δ works. For example, take $\delta = 1$.

   (b) Again, any δ works. For example, take $\delta = 1$.

   (c) Let $\delta = \epsilon$.

   (d) Let $\delta = \epsilon$.

   (e) Let $\epsilon > 0$. Let $\delta = \min\{1, \epsilon/5\}$. First suppose that $\epsilon \geq 5$ so that $\delta = 1$. If $|x - 2| < 1$, then $1 < x < 3$ so $1 < x^2 < 9$. It follows that $|x^2 - 4| < 5 = \epsilon$.

   Now suppose that $\epsilon < 5$ so that $\delta = \epsilon/5$. Since $|x - 2| < 1$, by the triangle inequality

   $$|x + 2| = |(x + 2) - 4| \leq |x - 2| + 4 \leq 5.$$

   Therefore

   $$|x^2 - 4| = |x + 2|\,|x - 2| \leq 5\,|x - 2| < 5\delta = \epsilon.$$

   (f) Let $\delta = \min\{1, \epsilon/(1 + 2a)\}$. Proceed as in (e), swapping 4 for a^2.

3. $\lim\limits_{\mathbf{x \to 1}} \dfrac{\mathbf{1}}{\mathbf{x}} = \mathbf{1}$

   In this exercise we show $\lim\limits_{x \to 1} \dfrac{1}{x} = 1$.

   This is a standard exercise in Real Analysis. You can find an excellent write up on this exercise at: *https://math.stackexchange.com/questions/418961/epsilon-delta-proof-that-lim-limits-x-to-1-frac1x-1*

   (a)

   $$\begin{aligned} \left|\frac{1}{x} - 1\right| &= \left|\frac{1}{x} - \frac{x}{x}\right| \\ &= \left|\frac{1 - x}{x}\right| \\ &= \left|\frac{x - 1}{x}\right| \\ &= \left|\frac{1}{x}\right| |x - 1| \end{aligned}$$

   (b)

   $$\begin{aligned} -\frac{1}{2} < x - 1 &< \frac{1}{2} \\ \frac{1}{2} < x &< \frac{3}{2} \\ \frac{2}{3} < \frac{1}{x} &< 2 \\ \left|\frac{1}{x}\right| &< 2 \end{aligned}$$

(c) Using Parts (a) and (b)

$$\left|\frac{1}{x}-1\right| = \left|\frac{1}{x}\right||x-1| < 2|x-1|$$

(d) Let $\epsilon > 0$ and $\delta = \min\left\{\frac{1}{2}, \frac{\epsilon}{2}\right\}$. Suppose $0 < |x-1| < \delta$. Since $\delta \leq 1/2$, $|x-1| < 1/2$, and by part (c) we have

$$\left|\frac{1}{x}-1\right| < 2\,|x-1| < 2\delta.$$

But $\delta \leq \epsilon/2$, so we conclude that

$$\left|\frac{1}{x}-1\right| < 2 \cdot \frac{\epsilon}{2} = \epsilon.$$

Therefore, $\lim_{x\to 1} \frac{1}{x} = 1$.

4. **A Squeeze Theorem for Real Limits**

In Tutorial 5 we proved a Squeeze Theorem for Sequences. Here we prove an analogous theorem for real functions.

Theorem (Squeeze Theorem for Real Functions)**.** *Let $E \subset \mathbb{R}$, let a be a limit point of E, let $f : E \to \mathbb{R}$, $g : E \to \mathbb{R}$, $h : E \to \mathbb{R}$ and let $L \in \mathbb{R}$. If*

$$f(x) \leq g(x) \leq h(x)$$

for all $x \in E$, and

$$\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L$$

then

$$\lim_{x\to a} g(x) = L$$

Solution: Proceed as in the solution in Tutorial 5B. Let $\delta = \min\{\delta_f, \delta_h\}$.

5. CHALLENGE **Rational Indicator Function**

Let $g : \mathbb{R} \to \mathbb{R}$ be defined by

$$g(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

In this exercise we prove $g(x)$ diverges as x approaches $\sqrt{2}$.

We proceed by contradiction. Assume there exists $L \in \mathbb{R}$ so that $\lim_{x\to\sqrt{2}} g(x) = L$. Let $\epsilon = \frac{1}{2}$. By definition, there exists $\delta > 0$ so that $|g(x) - L| < \epsilon$ whenever x satisfies $0 < |x - \sqrt{2}| < \delta$.

(a) Since there exists a rational number between any two real numbers, there exists $x_1 \in \mathbb{Q}$ so that

$$\sqrt{2} - \delta < x_1 < \sqrt{2} + \delta.$$

That is, $|x_1 - \sqrt{2}| < \delta$. Since $\sqrt{2}$ is irrational, $x_1 \neq \sqrt{2}$. Therefore, $0 < |x_1 - \sqrt{2}| < \delta$.

(b) Since $x_1 \in \mathbb{Q}$, $g(x_1) = 1$. Hence

$$|L-1| = |1-L| = |g(x_1) - L| < \epsilon = \tfrac{1}{2}.$$

Therefore $L > 1 - \frac{1}{2} = \frac{1}{2}$.

(c) Since there exists a rational number between any two real numbers, there exists $r \in \mathbb{Q}$ so that $0 < r < \delta$. Hence $0 < |r| < \delta$. Let $x_2 = r + \sqrt{2}$. Since r is rational and $\sqrt{2}$ is irrational, $x_2 = r + \sqrt{2}$ is irrational. Since $0 < |r| < \delta$, $0 < |x_2 - \sqrt{2}| = |r| < \delta$. Therefore, x_2 is an irrational number such that $0 < |x_2 - \sqrt{2}| < \delta$.

(d) Since $x_2 \in \mathbb{R} \setminus \mathbb{Q}$, $g(x_2) = 0$. Hence

$$|L| = |0 - L| = |g(x_1) - L| < \epsilon = \tfrac{1}{2}.$$

Therefore $L < 0 + \frac{1}{2} = \frac{1}{2}$.